



Pedestrian Heat Stress: Mapping Urban Heat Heterogeneity in Richmond's Jackson Ward Neighborhood

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Introduction

- Urban heat refers to highly-industrialized areas experiencing significantly higher temperatures than surrounding regions (Tong *et al.*, 2021).
- Urban heat is unevenly distributed, resulting in the formation of microclimates that differ from the climate of the overall surrounding region (Kottner *et al.*, 2018).
- Microclimates and urban heat are altered by surface materials, shade presence and other factors in the built environment (Lookingbill *et al.*, 2025). These factors greatly influence how people feel and experience heat at the pedestrian street level.
- Both wet bulb globe temperature (WBGT) and surface temperature (ST) can be utilized to assess heat stress and microclimate variability (Clark & Konrad, 2024).
- The general public's understanding of actual ambient temperature is arguably guided by weather station readings, of which may not be truly representative of the various conditions in the microenvironments that exist at the intra-neighborhood scale.
- Urban heat in Jackson Ward, a historically black neighborhood in Richmond, Virginia, has been shaped by discriminatory redlining, divisive I-95 highway construction and extensive urban renewal (Hoffman *et al.*, 2020).
- Mapping sidewalk temperatures provides insight into the infrastructure shaping urban heat issues experienced in Jackson Ward.
- The identification of the most vulnerable hotspots for pedestrians, particularly during heightened heat exposure in the summer, can inform public health cooling interventions for long-term heat mitigation in the neighborhood.

Research Questions

How do intra-neighborhood differences in urban heat manifest in Richmond's Jackson Ward neighborhood?

- Which areas of the neighborhood experience relatively higher and lower pedestrian heat stress (measured by WBGT)?
- What is the impact of structural shade and tree canopy shade on WBGT and ST on sidewalks?
- How much does street-level heat stress vary from the public's conventional understanding of heat informed by regional forecasts?

Methodology

Equipment and Devices

- Two teams with at least two researchers each
 - One researcher using a Kestrel 5400 Heat Stress Tracker
 - Measuring ambient WBGT
 - Tripod mounted with wind vane
 - One researcher using a FLIR One Edge Thermal Camera
 - Measuring the ST of the sidewalk



Figure 1. FLIR Thermal Camera (left) and Kestrel 5400 Heat Stress Tracker (right).

Data Collection Method

- Two pre-planned routes with target stops at 85 m intervals
 - Covering the entirety of Jackson Ward
- Walks along these routes took place on four separate days
 - 12:00pm to 3:00pm to capture peak heat
 - With a post-collection time adjustment
 - June 11th, 20th, 21st, and 24th

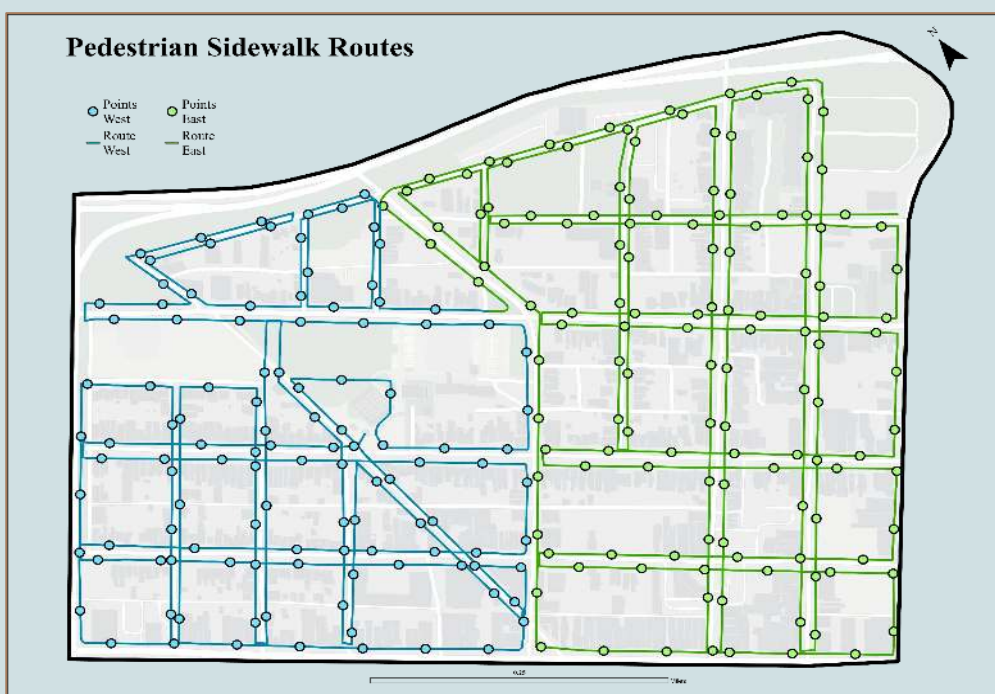


Figure 2. Sidewalk Data Collection Routes.

Interpolation

- Collected data points were converted to a continuous surface using Inverse Distance Weighting Method (IDW)
 - Reflects the idea that nearby points are more similar
 - Easy to use and gives quick, reasonable estimates

$$\hat{v}_1 = \frac{\sum_{i=1}^n \frac{1}{d^p_i} v_i}{\sum_{i=1}^n \frac{1}{d^p_i}}$$

Figure 3. Inverse Distance Weighting (IDW) Formula.

Mapping Pedestrian Heat Stress

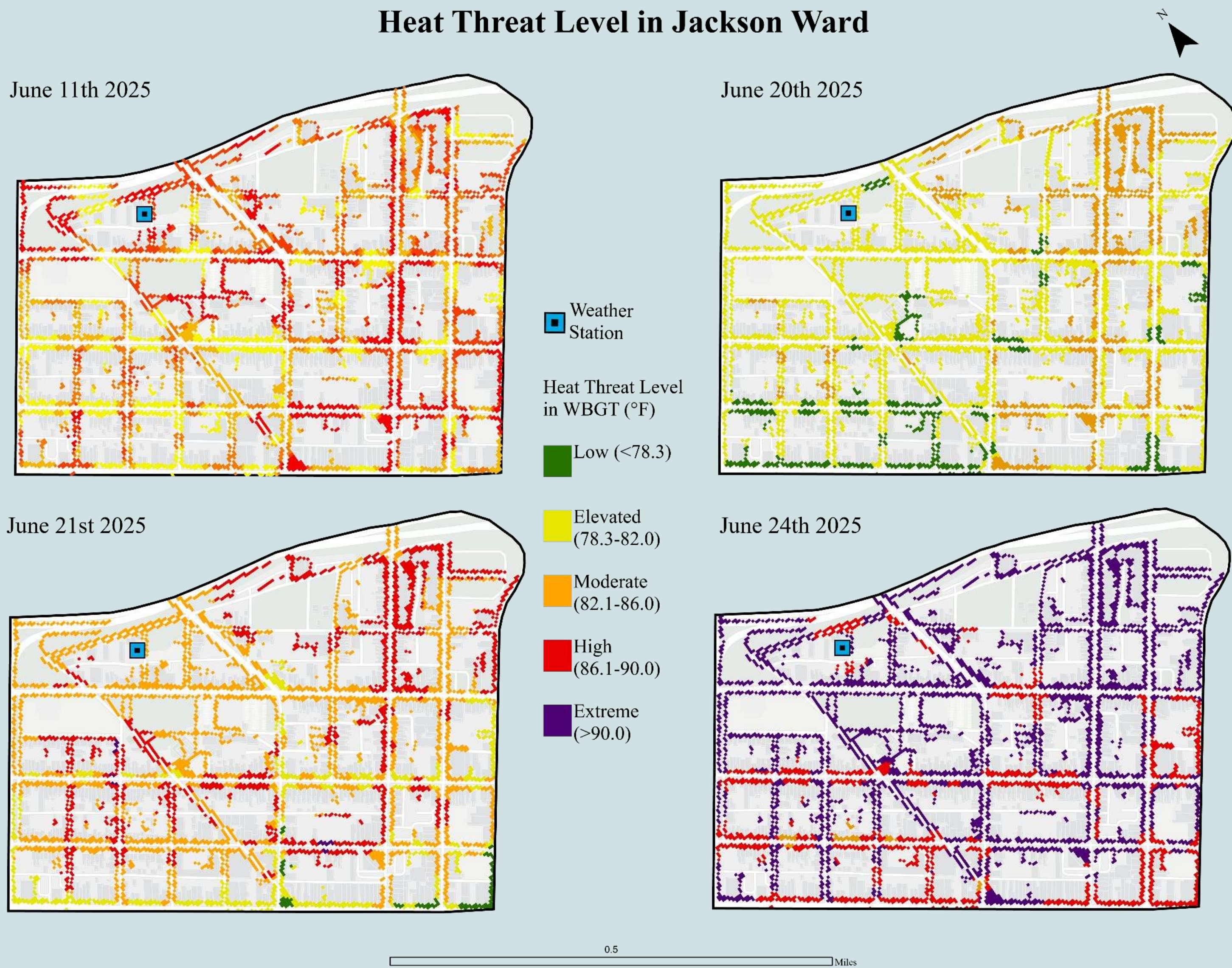


Figure 4. Map of Heat Threat Levels across Sidewalks of Jackson Ward in Richmond, Virginia.

- No street shows one uniform heat threat level on any day, with as many as four different levels recorded on one block
- WBGT peaks on June 24th, with majority of the neighborhood falling under the "high" or "extreme" heat designation
- Aggregating this data allowed us to identify relative "hot spots" in the neighborhood as priority areas for tree planting

Street-level Variation from Weather Station Measurements

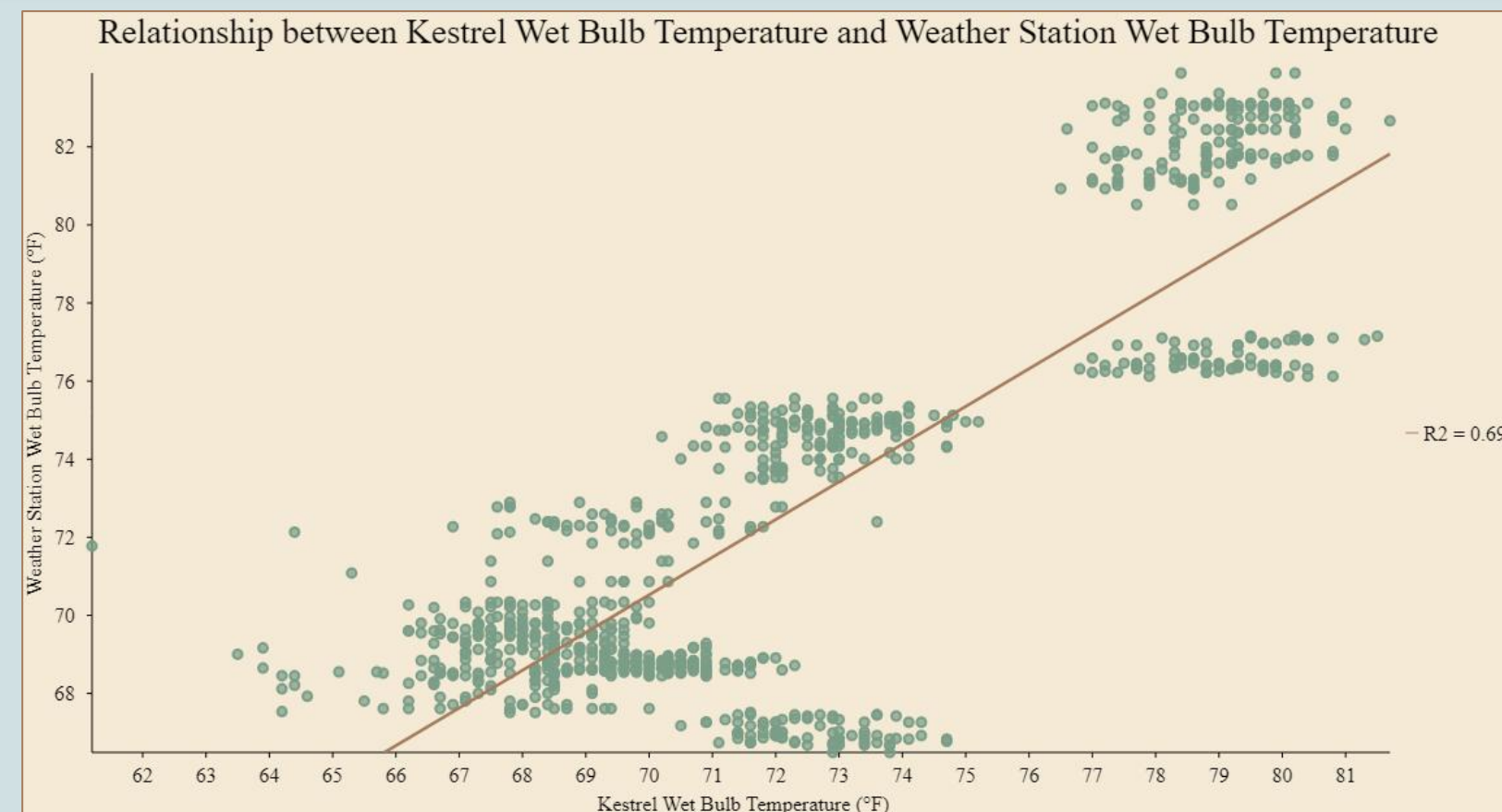


Figure 7. Scatterplot of Kestrel Wet Bulb Temperature Measurements and Weather Station Wet Bulb Temperature Measurements.

Comparison of Wet Bulb Temperature Measurements recorded by the Kestrels and the Neighborhood Weather Station	
Paired t-test p-value	<0.00
Standard Deviation of Wet Bulb Temperature Difference	±2.8°F
Range of Wet Bulb Temperature Difference	18.5°F

Table 3. Variance Statistics of Kestrel and Weather Station Wet Bulb Temperature Measurements.

- The results of the paired t-test indicate that there is a statistically significant difference between the Kestrel and Weather Station measurements of Wet Bulb temperature
- The standard deviation (±2.8°F) and range (18.5°F) of the difference indicate detectable differences for pedestrians at street level
- This variation speaks to a disconnect between the public's understanding of heat forecasts and street-level impacts

Contrasting Thermal Streetscapes

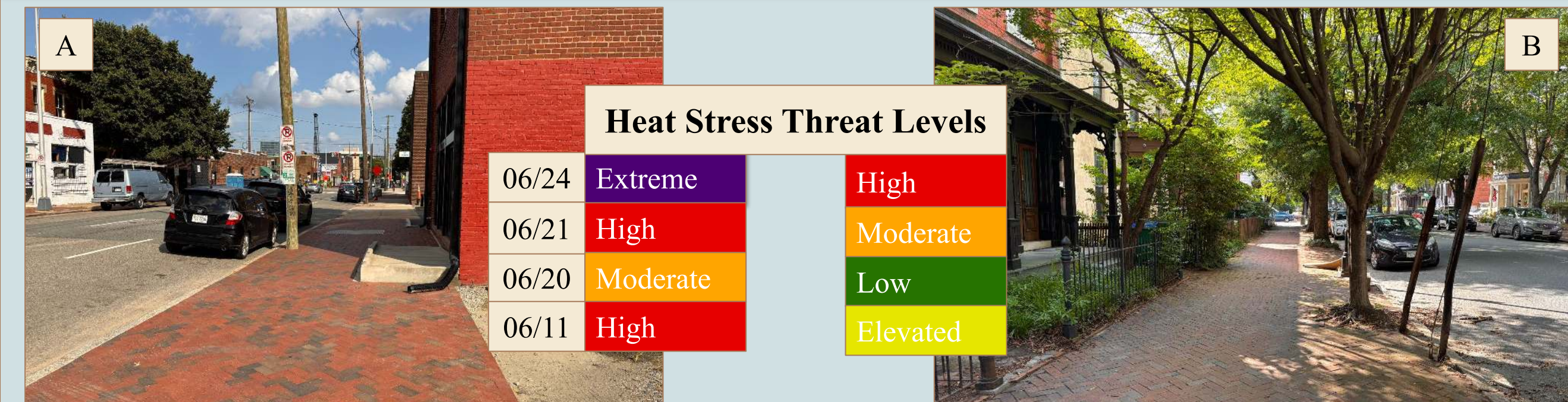


Figure 8. Hottest Identified Sidewalk (A) and Coolest Identified Sidewalk (B) in Jackson Ward. The hottest spot is located on W Marshall St. between 1st St. and N Adams St. The coolest spot is also located on W Marshall St., between N Monroe St. and N Madison St. The "Heat Stress Threat Levels" graphic details the WBGT heat stress designations (Figure 4) of each location on each day of data collection.

- On average, the hottest spot (A) had a 2.5°F hotter WBGT and a 44.7°F hotter ST than the coolest spot (B)
- These spots are located only four blocks (~0.25 miles) away from each other on the same street
- This underscores that pedestrian heat risk can be highly variable on the intra-neighborhood scale
- The stark contrast in urban tree canopy between the two locations aligns with our conclusions from the statistical analysis on shade types

Discussion and Conclusion

- Our findings demonstrate that within a neighborhood identified as an urban heat island, like Jackson Ward, there can be statistically significant heterogeneity in heat distribution
 - Urban heat varies with the built environment, and urban tree canopy was demonstrated to be a statistically significant determinant of sidewalk WBGT and ST
- As such, a tree planting initiative in Jackson Ward will be undertaken in Fall 2025 to reduce pedestrian heat stress in the hottest identified areas
- Moving forward, urban planners, government officials, and community leaders must contend with the effects of global warming by designing urban environments through the lens of heat mitigation
 - Furthermore, a more robust understanding of heat stress and the variation in street-level heat ought to be promoted to the public, as regional weather forecasts do not always sufficiently convey the variable extent of heat risk in urban areas.

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References

